

MULTIFUNCTION UNIT CONVERTER AND CALCULATOR USING MATLAB

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We often have to deal with measurements and mathematical calculations in our daily lives. Sometimes these require conversion of a particular unit of measurement into another unit. But it's not always possible to remember all those factors to be multiplied or divided to get the required unit. One solution to this problem is to use a unit converter device.

Here we present a unit converter and calculator program designed in MATLAB platform. This program comes handy if you have a PC, and is designed to be user-friendly with a graphical user interface (GUI).

The program can be used for multifunction unit conversion and simple calculations involving basic mathematical operations. The program output is shown in Fig. 1.

Software

The program is developed in MATLAB version 7—an easy and fast tool for creating a GUI program using various available inbuilt components. It generates '.m' file for the project automatically. To install the complete MATLAB software, you need a good system speed and sufficient memory space. System requirements depend on the MATLAB toolboxes specific to your application. You may check the exact details of system requirements from www.mathworks.com website.

The program files include:

1. **converter.fig**. It is the figure file that contains the GUI layout of the unit converter and calculator.

2. **converter_main.c**. It is the 'C' code for the entire software program. It makes the software a standalone application that can run the executable file without MATLAB being installed. This file is generated by the compiler.

3. **converter_data.c**. This file is also generated by MATLAB compiler. It is used to work with the executable file.

4. **converter.m**. This file is converted into a standalone application using `mcc` command in MATLAB. If MATLAB component runtime (MCR) installer is installed in your system, you can run the converter program without the complete MATLAB software. In this case, no special feature is required in your system to run the software. You just need Windows operating system.

In GUI, 'callback' is the most important function of each graphics component. The buttons for various operations are accessed through this function. The function makes various buttons show pop-up menus when clicked.

While coding the program, all the standard factors for conversions are

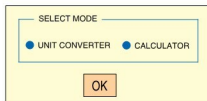


Fig. 1: Initial program output

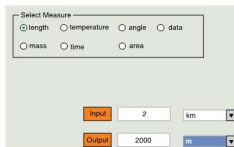


Fig. 2: Unit converter

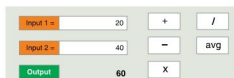


Fig. 3: Calculator



used. Some of these are:

- 1 km=1000 metres
- 1 hour=60 minutes
- 1 kg=2.67923 pounds
- 1 byte=8 bits
- 1 metre=3.280 feet
- 1 radian=57.2958 degrees
- 1 celsius=33.8 Fahrenheit

Operation

First, install the MATLAB software in your PC. Run the converter program by selecting the `converter.m` file from 'Debug' menu option in MATLAB. When the program (`converter.m`) is run, it prompts you to select the mode of operation—either unit converter or calculator.

The unit converter can convert units of length, time, temperature, area, mass, angle and computer data into other units (refer Fig. 2). Here you have to input the parameter (say, length) and select the quantity (say, 2) and its unit (say, km) followed by the output unit (say, m). The program will automatically output the quantity in the required unit (as 2000).

In calculator mode, the program prompts you for two inputs (refer Fig. 3). You need to fill in the edit box corresponding to these inputs. Click operation symbols such as '+' for addition, '-' for subtraction, 'x' for multiplication, '/' for division, and 'avg' for average operations. The result is displayed in the output box.

EFY note. All the relevant source codes of this article are available on www.efymag.com website. ●

The author is a final-year student of B.Tech in electronics and communications engineering with passion for C, C++ and MATLAB programming